第 17 章　多元函数积分学的预备知识

1.【答案】(A)

【解析】曲线 $y=y(x)$ 上任一点 $P(x,y)$ 处的切线方向为 $(\mathrm{d}x,\mathrm{d}y)$，且 $f(x,y)$ 变化率最大的方向为

$$\mathbf{grad}\, f=\frac{\partial f}{\partial x}\boldsymbol{i}+\frac{\partial f}{\partial y}\boldsymbol{j}=\mathrm{e}^{-(x^2+2y^2)}\cdot(-2x\boldsymbol{i}-4y\boldsymbol{j}).$$

由题意，得 $\dfrac{\mathrm{d}x}{-2x}=\dfrac{\mathrm{d}y}{-4y}$，则 $\ln|y|=\ln C_1+\ln x^2$，解得 $y=Cx^2$．又 $y(1)=2$，即 $C=2$，于是 $y=2x^2$．

2.【解析】设曲面上的点为 (x,y,z)，它到平面 $x+y-4z-1=0$ 的距离为 $d=\dfrac{|x+y-4z-1|}{\sqrt{1^2+1^2+(-4)^2}}$．

在约束条件 $3x^2-2xy+3y^2-4z=0$ 上求 d 的最小值等价于求 $S=(x+y-4z-1)^2$ 在上述约束条件下的最小值．

构造辅助函数 $F(x,y,z,\lambda)=(x+y-4z-1)^2+\lambda(3x^2-2xy+3y^2-4z)$，令

$$\begin{cases}\dfrac{\partial F}{\partial x}=2(x+y-4z-1)+6\lambda x-2\lambda y=0,\\ \dfrac{\partial F}{\partial y}=2(x+y-4z-1)-2\lambda x+6\lambda y=0,\\ \dfrac{\partial F}{\partial z}=-8(x+y-4z-1)-4\lambda=0,\\ \dfrac{\partial F}{\partial \lambda}=3x^2-2xy+3y^2-4z=0,\end{cases}$$

解得 $x=\dfrac{1}{4}$，$y=\dfrac{1}{4}$，$z=\dfrac{1}{16}$．根据几何意义，此点到平面 $x+y-4z-1=0$ 的距离最小，即

$$d_{\min}=\frac{\left|\frac{1}{4}+\frac{1}{4}-\frac{1}{4}-1\right|}{\sqrt{1^2+1^2+(-4)^2}}=\frac{\frac{3}{4}}{3\sqrt{2}}=\frac{\sqrt{2}}{8}.$$

3.【答案】$\dfrac{x-0}{1}=\dfrac{y-1}{\frac{1}{2}}=\dfrac{z-1}{1}$

【解析】将 L 写成以 z 为参数的方程，则 $L:\begin{cases}x=2(\sqrt{z}-1),\\ y=\sqrt{z},\\ z=z,\end{cases}$ 于是 $\begin{cases}x'_z=\dfrac{1}{\sqrt{z}},\\ y'_z=\dfrac{1}{2\sqrt{z}},\\ z'_z=1,\end{cases}$ 当 $y=1$ 时，$x=0$，

配套课程公众号[研大]
联系微信19316378888

$z=1$，于是该点处切线的方向向量为$\left(1,\frac{1}{2},1\right)$，故切线方程为$\frac{x-0}{1}=\frac{y-1}{\frac{1}{2}}=\frac{z-1}{1}$.

4.【答案】$\frac{x-1}{1}=\frac{y-1}{-2}=\frac{z-1}{1}$

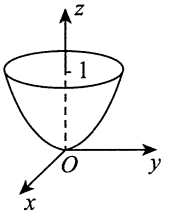
【解析】方程$\begin{cases}x^2+2y^2+3z^2=6,\\x+y+z=3\end{cases}$两边对$x$求导得$\begin{cases}2x+4y\frac{\mathrm{d}y}{\mathrm{d}x}+6z\frac{\mathrm{d}z}{\mathrm{d}x}=0,\\1+\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{\mathrm{d}z}{\mathrm{d}x}=0,\end{cases}$则

$$\begin{cases}4\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{(1,1,1)}+6\left.\frac{\mathrm{d}z}{\mathrm{d}x}\right|_{(1,1,1)}=-2,\\\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{(1,1,1)}+\left.\frac{\mathrm{d}z}{\mathrm{d}x}\right|_{(1,1,1)}=-1,\end{cases}$$

解得$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{(1,1,1)}=-2,\left.\frac{\mathrm{d}z}{\mathrm{d}x}\right|_{(1,1,1)}=1$.故曲线在点$(1,1,1)$处的切向量为$\boldsymbol{l}=\left(1,\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{(1,1,1)},\left.\frac{\mathrm{d}z}{\mathrm{d}x}\right|_{(1,1,1)}\right)=(1,-2,1)$.

于是，曲线在点$(1,1,1)$处的切线方程为$\frac{x-1}{1}=\frac{y-1}{-2}=\frac{z-1}{1}$.

5.【解析】空间图形Ω如图所示，可知$\bar{x}=\bar{y}=0$，



$$\bar{z}=\frac{\iiint\limits_{\Omega}z\mathrm{d}v}{\iiint\limits_{\Omega}\mathrm{d}v}=\frac{\iint\limits_{x^2+y^2\leqslant1}\mathrm{d}\sigma\int_{x^2+y^2}^{1}z\mathrm{d}z}{\iint\limits_{x^2+y^2\leqslant1}\mathrm{d}\sigma\int_{x^2+y^2}^{1}\mathrm{d}z}=\frac{\int_0^{2\pi}\mathrm{d}\theta\int_0^1\frac{1}{2}(1-r^4)r\mathrm{d}r}{\int_0^{2\pi}\mathrm{d}\theta\int_0^1(1-r^2)r\mathrm{d}r}=\frac{2}{3},$$

故形心为$P\left(0,0,\frac{2}{3}\right)$，因此抛物面$\Sigma$上任一点$(x,y,z)$到点$P$的距离为

$$d=\sqrt{(x-0)^2+(y-0)^2+\left(z-\frac{2}{3}\right)^2}=\sqrt{x^2+y^2+\left(z-\frac{2}{3}\right)^2}.$$

故问题转化为求$u=x^2+y^2+\left(z-\frac{2}{3}\right)^2$在约束条件$x^2+y^2=z$下的最小值，设

$$F(x,y,z,\lambda)=x^2+y^2+\left(z-\frac{2}{3}\right)^2+\lambda(x^2+y^2-z),$$

令

$$\begin{cases}F_x'=2x+2\lambda x=0,\\F_y'=2y+2\lambda y=0,\\F_z'=2\left(z-\frac{2}{3}\right)-\lambda=0,\\F_\lambda'=x^2+y^2-z=0,\end{cases}$$

配套课程公众号[研大]
联系微信19316378888

解得 $x^2+y^2=\frac{1}{6}$，$z=\frac{1}{6}$ 或 $x=y=z=0$．由题设可知最短距离必存在，且 $d\Big|_{\substack{x^2+y^2=\frac{1}{6}\\ z=\frac{1}{6}}}=\frac{\sqrt{15}}{6}<d\big|_{(0,0,0)}=\frac{2}{3}$，故 $d_{\min}=\frac{\sqrt{15}}{6}$．

6.【答案】 $x+3z-3f(0,0)=0$

【解析】 曲线 $\begin{cases}z=f(x,y),\\ y=0\end{cases}$ 可写成参数式：$\begin{cases}x=t,\\ y=0,\\ z=f(t,0),\end{cases}$ 则 $\boldsymbol{\tau}=(x_t',y_t',z_t')\big|_{t=0}=(1,0,f_x'(0,0))=(1,0,3)$，故所求法平面方程为 $x+3z-3f(0,0)=0$．

7.【答案】 $y=2$

【解析】 记 $F(x,y,z)=\mathrm{e}^z-xz+y-3$，则 $F_x'=-z$，$F_y'=1$，$F_z'=\mathrm{e}^z-x$，代入点 $(1,2,0)$，得

$$F_x'(1,2,0)=0,\ F_y'(1,2,0)=1,\ F_z'(1,2,0)=0,$$

所以曲面 $\mathrm{e}^z-xz+y=3$ 在点 $(1,2,0)$ 处的切平面方程为 $0\cdot(x-1)+1\cdot(y-2)+0\cdot(z-0)=0$，即 $y=2$．

8.【答案】 (B)

【解析】 函数 $f(u,v)$ 在点 $(1,2)$ 处的方向导数的最大值应等于 $f(u,v)$ 在点 $(1,2)$ 处梯度的模，即 $\sqrt{\left(\frac{\partial f}{\partial u}\Big|_{\substack{u=1\\v=2}}\right)^2+\left(\frac{\partial f}{\partial v}\Big|_{\substack{u=1\\v=2}}\right)^2}$．为此，先计算偏导数 $\frac{\partial f}{\partial u}\Big|_{\substack{u=1\\v=2}}$ 和 $\frac{\partial f}{\partial v}\Big|_{\substack{u=1\\v=2}}$．

令 $x-y=u$，$x+\mathrm{e}^y=v$，则当 $u=1$，$v=2$ 时，有 $x-y=1$，$x+\mathrm{e}^y=2$，解得 $x=1$，$y=0$．

$f(x-y,x+\mathrm{e}^y)=x^2-y^2$ 两边分别对 x，y 求偏导数，得 $\frac{\partial f}{\partial u}+\frac{\partial f}{\partial v}=2x$，$-\frac{\partial f}{\partial u}+\frac{\partial f}{\partial v}\mathrm{e}^y=-2y$，将 $x=1$，$y=0$ 代入上式，可解得 $\frac{\partial f}{\partial u}\Big|_{\substack{u=1\\v=2}}=\frac{\partial f}{\partial v}\Big|_{\substack{u=1\\v=2}}=1$．

因此，$f(u,v)$ 在点 $(1,2)$ 处的方向导数的最大值等于 $\sqrt{2}$．故选(B).

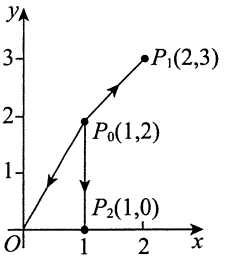
9.【答案】 $(0,\mathrm{e},2)$

【解析】 $\mathbf{grad}\,f(x,y,z)\big|_{(1,1,2)}=\left(2x\int_1^y\mathrm{e}^t\mathrm{d}t,x^2\mathrm{e}^y,2\right)\Big|_{(1,1,2)}=(0,\mathrm{e},2)$．

10.【答案】 $\sqrt{10}$

配套课程公众号[研大]
联系微信19316378888

【解析】如图所示，$\boldsymbol{l}_1=\overrightarrow{P_0P_1}=(1,1)$，$\boldsymbol{l}_2=\overrightarrow{P_0P_2}=(0,-2)$，且



$$\boldsymbol{l}_1^{\circ}=(\cos\alpha_1,\cos\beta_1)=\left(\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right),\ \boldsymbol{l}_2^{\circ}=(\cos\alpha_2,\cos\beta_2)=(0,-1).$$

由方向导数计算公式，有

$$\left.\frac{\partial f}{\partial x}\right|_{P_0}\cdot\frac{1}{\sqrt{2}}+\left.\frac{\partial f}{\partial y}\right|_{P_0}\cdot\frac{1}{\sqrt{2}}=2\sqrt{2},\ \left.\frac{\partial f}{\partial x}\right|_{P_0}\cdot 0+\left.\frac{\partial f}{\partial y}\right|_{P_0}\cdot(-1)=-3,$$

解得 $\left.\frac{\partial f}{\partial x}\right|_{P_0}=1$，$\left.\frac{\partial f}{\partial y}\right|_{P_0}=3$，故 z 在点 P_0 处的最大方向导数为

$$\left|\mathbf{grad}z\big|_{P_0}\right|=\sqrt{\left(\left.\frac{\partial f}{\partial x}\right|_{P_0}\right)^2+\left(\left.\frac{\partial f}{\partial y}\right|_{P_0}\right)^2}=\sqrt{10}.$$

【注】本题的问题可作如下推广：设 $z=f(x,y)$ 可微，记任意一点 $P_0(x_0,y_0)$，从 P_0 出发，沿两条不共线的方向 $\boldsymbol{l}_1^{\circ}=(\cos\alpha_1,\cos\beta_1)$ 与 $\boldsymbol{l}_2^{\circ}=(\cos\alpha_2,\cos\beta_2)$ 的方向导数分别为

$$\begin{cases}\left.\dfrac{\partial f}{\partial \boldsymbol{l}_1^{\circ}}\right|_{P_0}=\left.\dfrac{\partial f}{\partial x}\right|_{P_0}\cdot\cos\alpha_1+\left.\dfrac{\partial f}{\partial y}\right|_{P_0}\cdot\cos\beta_1,\\ \left.\dfrac{\partial f}{\partial \boldsymbol{l}_2^{\circ}}\right|_{P_0}=\left.\dfrac{\partial f}{\partial x}\right|_{P_0}\cdot\cos\alpha_2+\left.\dfrac{\partial f}{\partial y}\right|_{P_0}\cdot\cos\beta_2,\end{cases}$$

其中 $\begin{vmatrix}\cos\alpha_1 & \cos\beta_1\\ \cos\alpha_2 & \cos\beta_2\end{vmatrix}\neq 0$.

①若 $\left.\frac{\partial f}{\partial \boldsymbol{l}_1^{\circ}}\right|_{P_0}$，$\left.\frac{\partial f}{\partial \boldsymbol{l}_2^{\circ}}\right|_{P_0}$ 不全为 0，则该非齐次方程组有唯一解，如本题的解答过程.

②若 $\left.\frac{\partial f}{\partial \boldsymbol{l}_1^{\circ}}\right|_{P_0}=\left.\frac{\partial f}{\partial \boldsymbol{l}_2^{\circ}}\right|_{P_0}=0$，则该齐次方程组只有零解，即 $\left.\frac{\partial f}{\partial x}\right|_{P_0}=\left.\frac{\partial f}{\partial y}\right|_{P_0}=0$，故 $\mathrm{d}f\big|_{P_0}=\left.\frac{\partial f}{\partial x}\right|_{P_0}\mathrm{d}x+\left.\frac{\partial f}{\partial y}\right|_{P_0}\mathrm{d}y=0$. 由 P_0 的任意性，有 $\mathrm{d}f=0$，故 $f(x,y)$ 为一常数.

11.【答案】 $\sqrt{66}$

【解析】函数沿梯度方向的方向导数最大，且最大值就是梯度的模，又 $\mathbf{grad}\,u\big|_{(1,1,-2)}=(y,x,-4z)\big|_{(1,1,-2)}=(1,1,8)$，因此最大方向导数为 $\sqrt{1+1+8^2}=\sqrt{66}$.

12.【答案】 (B)

【解析】$\|\mathbf{grad}\,f(0,0)\|=-a$，又 $|[\boldsymbol{\alpha},\boldsymbol{\beta}]|=\|\boldsymbol{\alpha}\|\|\boldsymbol{\beta}\|\cdot|\cos\theta|\leqslant\|\boldsymbol{\alpha}\|\|\boldsymbol{\beta}\|$，其中 θ 为 $\boldsymbol{\alpha}$ 与 $\boldsymbol{\beta}$ 的夹角.

配套课程公众号[研大]
联系微信19316378888

于是$|[\mathbf{grad}\, f(0,0),(b,c)]| \leqslant \|\mathbf{grad}\, f(0,0)\| \cdot \sqrt{b^2+c^2} = -a\sqrt{b^2+c^2}$，故选(B).

13.【答案】(D)

【解析】$\mathbf{grad}\, f|_{p_0} \cdot \overrightarrow{p_0p} = \left(\left.\frac{\partial f}{\partial x}\right|_{p_0}, \left.\frac{\partial f}{\partial y}\right|_{p_0}\right)\begin{pmatrix} x-x_0 \\ y-y_0 \end{pmatrix} = \left.\frac{\partial f}{\partial x}\right|_{p_0} \cdot (x-x_0) + \left.\frac{\partial f}{\partial y}\right|_{p_0} \cdot (y-y_0)$，

$$|\overrightarrow{p_0p}| = \sqrt{(x-x_0)^2+(y-y_0)^2},$$

又由于$f(x,y)$在点(x_0,y_0)处可微，于是有

$$\lim_{p \to p_0} \frac{f(x,y)-f(x_0,y_0)-\mathbf{grad}\, f|_{p_0} \cdot \overrightarrow{p_0p}}{|\overrightarrow{p_0p}|} = 0.$$

14.【解析】曲面Σ在点$P\left(\frac{1}{2},-\frac{1}{2},\frac{1}{2}\right)$的外侧法向量为$\boldsymbol{n} = (2x,4y,2z)\left|_{\left(\frac{1}{2},-\frac{1}{2},\frac{1}{2}\right)}\right. = (1,-2,1)$，其方向余弦为$\cos\alpha = \frac{1}{\sqrt{6}}$，$\cos\beta = -\frac{2}{\sqrt{6}}$，$\cos\gamma = \frac{1}{\sqrt{6}}$.

函数f沿方向$\boldsymbol{n}$的方向导数为$\frac{\partial f}{\partial \boldsymbol{n}} = \frac{\partial f}{\partial x}\cos\alpha + \frac{\partial f}{\partial y}\cos\beta + \frac{\partial f}{\partial z}\cos\gamma = \frac{1}{\sqrt{6}}(4x-4y+1)$.

按照题意，即求函数$S = \frac{1}{\sqrt{6}}(4x-4y+1)$在条件$x^2+2y^2+z^2=1$下的最大值.

设$F(x,y,z,\lambda) = 4x-4y+1+\lambda(x^2+2y^2+z^2-1)$，则由方程组

$$\begin{cases} \frac{\partial F}{\partial x} = 4+2\lambda x = 0, \\ \frac{\partial F}{\partial y} = -4+4\lambda y = 0, \\ \frac{\partial F}{\partial z} = 2\lambda z = 0, \\ \frac{\partial F}{\partial \lambda} = x^2+2y^2+z^2-1 = 0, \end{cases}$$

解得驻点为$M_1\left(\frac{2}{\sqrt{6}},-\frac{1}{\sqrt{6}},0\right)$，$M_2\left(-\frac{2}{\sqrt{6}},\frac{1}{\sqrt{6}},0\right)$.

因最大值必存在，比较$\left.\frac{\partial f}{\partial \boldsymbol{n}}\right|_{M_1} = 2+\frac{1}{\sqrt{6}}$，$\left.\frac{\partial f}{\partial \boldsymbol{n}}\right|_{M_2} = -2+\frac{1}{\sqrt{6}}$的大小，可知$M_1\left(\frac{2}{\sqrt{6}},-\frac{1}{\sqrt{6}},0\right)$即为所求.

15.【答案】$\boldsymbol{i}+\sin 1\boldsymbol{k}$

【解析】记$\boldsymbol{F}(x,y,z) = (P,Q,R) = (|x|\cos y, -y\sin z, z)$，因为所求点为$(1,1,0)$，$x=1>0$，故$P = |x|\cos y = x\cos y$，则

配套课程公众号[研大]
联系微信19316378888

$$\mathbf{rot}\,\boldsymbol{F}(1,1,0)=\left|\begin{matrix}\boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k}\\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z}\\ P & Q & R\end{matrix}\right|_{(1,1,0)}=\left|\begin{matrix}\boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k}\\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z}\\ x\cos y & -y\sin z & z\end{matrix}\right|_{(1,1,0)}$$

$$=(y\cos z\boldsymbol{i}+x\sin y\boldsymbol{k})\big|_{(1,1,0)}=\boldsymbol{i}+\sin 1\boldsymbol{k}.$$

云上印
4.9元包邮@5分钱/每张
微信刷题小程序:酷酷刷题
免费刷肖1000徐涛腿姐等18位名师题库
支持错题导出、添加笔记、电脑端刷题...
配套课程公众号[研大]
联系微信19316378888